Lab 8, Act 2: Shrinking the Coefficients

What changes when all eight pond features compete in one model?

Answer key. Answers are in magenta sans-serif type; longer answers have a bar in the margin.

The scenario

In Act 1 you compared seven hand-picked models for Bd prevalence in the 60-pond survey. With eight candidate pond features there are $2^8 = 256$ possible subsets, and AIC couldn't separate temperature from elevation. Another approach is to keep all eight features in one model and **penalize** the size of their coefficients, so a feature keeps a large coefficient only if the data insist on it.

What happens to each pond feature's coefficient when all eight compete, and what can we tell the managers?

Try Part 1 on paper first, before you open a laptop or start typing code.

Part 1: Plan it on paper (~15 min)

1a. Write the ridge-penalized NLL for the logistic-binomial model with all eight standardized predictors, as a function of $\beta_0, \beta_1, \ldots, \beta_8$ and the penalty λ. Should β_0 be penalized? Why or why not?

$$\text{PNLL}(\beta) = -\sum_{i=1}^{60} [y_i \log p_i + (n_i - y_i) \log(1 - p_i)] + \lambda \sum_{j=1}^{8} \beta_j^2 + \text{const.}, \qquad \text{logit}(p_i) = \beta_0 + \sum_{j=1}^{8} \beta_j x_{ij}$$

No: β_0 sets the overall prevalence. Shrinking it toward 0 would pull every pond's prevalence toward 50% for no reason.

1b. Why standardize the predictors before penalizing them?

One penalty applies to every coefficient, but a raw coefficient's size depends on units (per meter of elevation vs. per °C). Standardizing puts every coefficient on the same per-SD scale, so the penalty treats them alike.

1c. In plain English (not R code), describe step by step how you would choose λ for your ridge model by 5-fold cross-validation, scoring each held-out fold by its NLL.

> Randomly split the 60 ponds into 5 groups of 12. For each candidate λ in a grid: for each group in turn, fit the ridge model to the other 48 ponds, then compute the (unpenalized) NLL of the 12 held-out ponds using those coefficients; add up the 5 held-out NLLs. Choose the λ with the smallest total.

Checkpoint: finalize your plan (~5 min)

Look back over 1a-1c and fix anything that no longer looks right before you start coding.

Part 2: Fit it (laptop, ~40 min)

Using AI: each question has a recommendation. *Recommended: no AI* marks the core skill of the lab (writing a penalized likelihood and fitting it); R's help pages, web search, and classmates are fine. *AI optional* marks work where the results matter but writing the code by hand isn't the point. After each question, circle how you actually used AI.

What to turn in: your filled-in answers and your 2b figure as an image (upload in WebCampus). You don't need to submit your code.

The data are in `BdRegionalSurvey.csv` on the course website: https://kevintshoemaker.github.io/NRES-746/data/BdRegionalSurvey.csv. Read it and standardize the eight predictors (columns 2 to 9):

```
data_url <-
  "https://kevintshoemaker.github.io/NRES-746/data/BdRegionalSurvey.csv"
ponds <- read.csv(data_url)
X <- scale(as.matrix(ponds[, 2:9]))
```

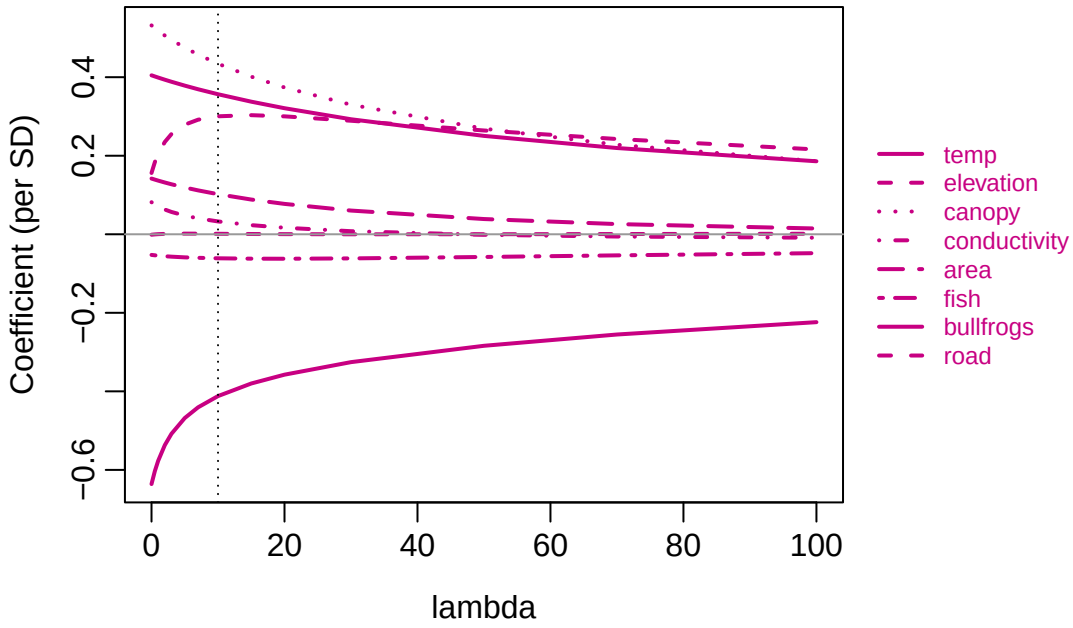
2a. *(Recommended: no AI.)* Write a function for your penalized NLL from 1a and fit it with `optim(..., method = "BFGS")` at $\lambda = 0$ and $\lambda = 10$. Check: the temperature coefficient should be about -0.64 at $\lambda = 0$ (the same as `glm()`) and -0.41 at $\lambda = 10$. Fill in those two columns of the table.

Predictor	MLE (lambda = 0)	Ridge (lambda = 10)	Lasso (2c)
water temp.	−0.64	−0.41	*−0.40*
elevation	*0.16*	*0.30*	*0.17*
canopy	*0.53*	*0.43*	*0.31*
conductivity	*0.08*	*0.03*	*0.00*
area	*0.14*	*0.10*	*0.00*
fish	*−0.05*	*−0.06*	*0.00*
bullfrogs	*0.40*	*0.36*	*0.26*
road distance	*0.00*	*0.00*	*0.00*

```
ridge_nll <- function(par, lambda) {
  p <- plogis(par[1] + X %*% par[-1])
  -sum(dbinom(ponds$bd_positive, ponds$frogs_swabbed, p, log = TRUE)) +
    lambda * sum(par[-1]^2)
}
optim(rep(0, 9), ridge_nll, lambda = 10, method = "BFGS")$par
```

AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

2b. *(AI optional.)* Fit your ridge model at a grid of λ values from 0 to 100 (say 0, 1, 2, 5, 10, 20, 50, 100), and record all eight slope estimates at each one. Then make a **ridge path**: one plot with λ on the x-axis, the coefficient estimate on the y-axis, and one line per predictor, so you can follow how each estimate changes as the penalty grows. Mark $\lambda = 10$. Turn in this figure.



AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

2c. *(AI optional.)* Fit the lasso with the `glmnet` package: `cv.glmnet()` with `family = "binomial"` and `nfolds = 10`, run right after `set.seed(1)`. For binomial counts, `glmnet` expects the response as `cbind(failures, successes)`. Fill in the lasso column with the coefficients at `lambda.1se` (write 0 for any it dropped).

```
library(glmnet)
y <- cbind(ponds$frogs_swabbed - ponds$bd_positive, ponds$bd_positive)
set.seed(1)
cv_lasso <- cv.glmnet(X, y, family = "binomial", nfolds = 10)
coef(cv_lasso, s = "lambda.1se")
```

AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

2d. Where did you use AI, if anywhere, and what did you have to check or fix? If you didn't use it, what was hardest to get working?

Open-ended.

Part 3: Interpret (~20 min)

3a. Temperature and elevation are strongly correlated ($r = -0.94$): on average, a pond 1 SD warmer than average is also about 0.94 SD lower. Here are two ways to divide their shared effect:

- Option A: $\beta_{temp} = -0.6$, $\beta_{elev} = 0$
- Option B: $\beta_{temp} = -0.3$, $\beta_{elev} = 0.3$

For a pond 1 SD warmer and 0.94 SD lower than average, how much does each option change the log-odds of infection? What does each option add to the ridge penalty ($\beta_{temp}^2 + \beta_{elev}^2$)? Use your answers to explain what the elevation line does in your 2b figure, and why the other six predictors behave differently.

> Change in log-odds: A gives -0.60; B gives $-0.3 + 0.3(-0.94)$ = -0.58, almost the same.
>
> Ridge penalty: A adds 0.36, B adds 0.18, so ridge prefers the split.
>
> In the 2b figure, elevation grows from 0.16 to 0.30 (at $\lambda = 10$) while temperature shrinks from -0.64 to -0.41: splitting the shared effect is cheaper, so ridge moves weight onto elevation. By $\lambda = 100$ they are about equal in size (-0.22 and 0.22). The other six have no near-duplicate, so the penalty only shrinks them.

3b. Compute the lasso penalty, $|\beta_{temp}| + |\beta_{elev}|$, for Options A and B. Does the lasso reward splitting the way ridge does? Your lasso kept both predictors. The table shows `glm()` fits with and without elevation. Why does adding elevation triple temperature's standard error, and how much would you trust either method's split between the two?

Model	Temperature (SE)	Elevation (SE)
temp + bullfrogs + canopy	-0.74 (0.08)	not in model
temp + elevation + bullfrogs + canopy	-0.54 (0.24)	0.21 (0.24)

> Lasso penalty: 0.60 for both, so the lasso doesn't reward splitting (here it kept temperature -0.40 and elevation 0.17).
>
> The two predictors carry nearly the same information, so the data determine their combined effect well but not how to divide it; many splits fit about equally well. Neither method's split should be trusted: it reflects the penalty, not the ponds. Ridge splits evenly because of its squared penalty, and the lasso goes wherever the data lean slightly. Neither tells us whether cold water or something else about high elevation drives Bd.

Reflection (~10 min)

The managers asked which pond features predict where Bd is common. Write two or three sentences they could act on, including what these data can and can't tell them about temperature versus elevation.

Also record any changes you made, new insights, and any "muddiest points" or topics you want to review.

> Sample: Bd is more common in colder ponds, in ponds with bullfrogs, and in more shaded ponds, so those are the ones to prioritize for monitoring. These data can't say whether cold water itself or something else about high elevation is responsible, because the two go together; for choosing ponds, either one works.

If you finish early. *(Recommended: no AI.)* Code your cross-validation plan from 1c and use it to choose λ for your ridge model from a grid of values. Which λ did you pick?

$\lambda =$ 10 from the grid 0, 1, 2, 5, 10, 20, 50, 100, with folds assigned after 'set.seed(1)'; your choice depends on your folds and grid.

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